

## Data Mining – Lecture 10 – Clustering

1- What is the primary goal of cluster analysis?

- a- Classifying data objects into distinct categories
- b- Identifying outliers within the dataset
- c- Discovering previously unknown groups within the data
- d- Determining the similarity between data objects

Answer : c.

2- In cluster analysis, what do clusters represent?

- a- Disjointed data objects
- b- Similarity between data objects
- c- Explicit classes
- d- Previously known groups within the data

Answer : c.

3- Which of the following applications can benefit from cluster analysis?

- a- Financial forecasting
- b- Agricultural yield prediction
- c- Web search optimization
- d- All of the above

Answer : d.

4- How can clustering be utilized besides discovering groups within the data?

- a- Predicting future trends
- b- Pre-processing data
- c- Filtering noisy data
- d- Generating reports

Answer : b.

5- In cluster analysis, objects within a cluster are typically similar to each other but dissimilar to objects in other clusters.

- a- True
- b- False

Answer : a.

6- Which cluster analysis method is effective for small- to medium-size datasets and finds mutually exclusive clusters of spherical shape?

- a- Hierarchical methods
- b- Density-based methods
- c- Grid-based methods
- d- Partitioning methods

Answer : d.

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7- Which cluster analysis method involves clustering in a hierarchy involving multiple levels and may consider object linkages?

- a- Partitioning methods
- b- Hierarchical methods
- c- Density-based methods
- d- Grid-based methods

Answer : b .

8- Density-based methods in cluster analysis can:

- a- Find arbitrarily shaped clusters
- b- Correct erroneous merges/splits
- c- Use a multi-resolution grid data structure
- d- Employ a distance-based approach

Answer : a.

9- Clusters are represented by their centroids .

- a- True
- b- False

Answer : a. Note “ Centroid is a cluster’s center point “

10- In k-means centroid is the \_\_\_ of points within cluster.

- a- Mean
- b- Mode
- c- Both
- d- None

Answer : a. note: “ we divide dataset into k mutually exclusive clusters”

11- What is the centroid of cluster C2 in the initial clustering?

- a- (6, 17/3)
- b- (51/3, 6)
- c- (6, 7)
- d- (7, 6)

Answer : a. Note ( “ most of the questions will be like this “)

12- X1 will belong to \_\_\_\_\_ after the first table of k means .

- a- C1
- b- C2
- c- C3
- d- None

Answer : c.

**Cluster the eight points in table using k-means.** Assume that  $k = 3$  and that initially the points are assigned to clusters as follows:  $C1 = \{x1, x2, x3\}$ ,  $C2 = \{x4, x5, x6\}$ ,  $C3 = \{x7, x8\}$ .

• Apply the k-means algorithm until convergence (i.e., until the clusters do not change), using the Manhattan distance.

(Hint: The Manhattan distance is:  $d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{in} - x_{jn}|$ .) Make sure you clearly identify the final clustering and show your steps.

|    | A1 | A2 |
|----|----|----|
| x1 | 2  | 10 |
| x2 | 2  | 5  |
| x3 | 8  | 4  |
| x4 | 5  | 8  |
| x5 | 7  | 5  |
| x6 | 6  | 4  |
| x7 | 1  | 2  |
| x8 | 4  | 9  |

## Data Mining – Lecture 10 – Clustering

12- What is the primary difference between agglomerative and divisive clustering?

- a- Agglomerative clustering is bottom-up, while divisive clustering is top-down.
- b- Agglomerative clustering is top-down, while divisive clustering is bottom-up.
- c- Agglomerative clustering merges clusters, while divisive clustering divides clusters.
- d- Agglomerative clustering divides clusters, while divisive clustering merges clusters.

Answer : a.

13- In agglomerative clustering, what happens in each iteration?

- a- Clusters are merged until a single cluster is formed.
- b- Clusters are divided until each object has its own cluster.
- c- Clusters are merged until termination condition or a single cluster is formed.
- d- Clusters are divided until all objects are similar enough.

Answer : c.

14- How does divisive clustering terminate?

- a- When all objects have their own cluster.
- b- When a single cluster is formed.
- c- When objects in clusters are similar "enough".
- d- A and C

Answer : d.

15- What is the primary goal of DBSCAN?

- a- Finding clusters with spherical shapes
- b- Identifying outliers within the dataset
- c- Discovering dense clusters based on neighborhood density
- d- Partitioning a set of data objects into mutually exclusive clusters

Answer : c.

16- In DBSCAN, what defines a core object?

- a- An object located at the center of a cluster
- b- An object with a large number of connections to other objects
- c- An object whose  $\epsilon$ -neighborhood contains at least MinPts objects
- d- An object with the highest density in the dataset

Answer : c.

17- How is the neighborhood density in DBSCAN defined?

- a- The maximum distance between two objects
- b- The number of objects within a specified radius of a given object
- c- The average distance between all objects in the dataset
- d- The number of dimensions in the dataset

Answer : b.

## Data Mining – Lecture 10 – Clustering

18- When is an object  $p$  considered directly density-reachable from another object  $q$  in DBSCAN?

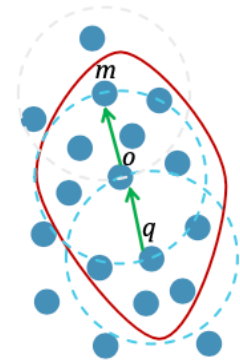
- a- When  $q$  is a core object and  $p$  is in the  $\epsilon$ -neighborhood of  $q$
- b- When  $p$  is a core object and  $q$  is in the  $\epsilon$ -neighborhood of  $p$
- c- When both  $p$  and  $q$  are outliers in the dataset
- d- When  $p$  and  $q$  are in the same cluster

Answer : a.

19- In DBSCAN, when are two objects  $q$  and  $m$  considered density-connected?

- a- When they have the same coordinates
- b- When they are directly density-reachable from each other
- c- When there is an object  $o$  such that both  $q$  and  $m$  are density-reachable from  $o$
- d- When they belong to the same cluster

Answer : c.



20- What is the purpose of the Hopkins Statistic in assessing clustering tendency?

- a- To measure the compactness of clusters
- b- To determine whether a dataset has a non-random structure
- c- To calculate the silhouette coefficient
- d- To evaluate the homogeneity of clusters

Answer : b.

21- How is the Hopkins Statistic calculated?

- a- By finding the nearest neighbors of each point in the dataset
- b- By computing the average distance between objects in different clusters
- c- By comparing the clustering against ground truth
- d- By comparing the distances of sampled points to their nearest neighbors

Answer : d.

22- DBSCAN algorithm is a density based clustering algorithm.

- a- True
- b- False

Answer : a.

23- What does a Hopkins Statistic value close to 0.5 indicate?

- a- The dataset has a non-random structure
- b- The dataset is uniformly distributed
- c- The clustering is of high quality
- d- The dataset has a random structure

Answer : b.

## Data Mining – Lecture 10 – Clustering

24- Which type of evaluation method compares clustering against ground truth?

- a- Extrinsic methods
- b- Intrinsic methods
- c- Supervised methods
- d- Unsupervised methods

Answer : a.

25- In the context of evaluation, what does "precision" measure?

- a- How many objects in the same cluster share the same category
- b- How many objects of the same category are assigned to the same cluster
- c- How well-separated clusters are from each other
- d- How pure each cluster is

Answer : a. note “ b is recall “

26- What does a positive silhouette coefficient indicate?

- a- The clustering is good
- b- The clustering is bad
- c- The dataset has a non-random structure
- d- The dataset is uniformly distributed

Answer : a.

" شكرا لكم ولا تنسونا بالدعاء "